

Solution to Exercise 12 — Analysis on Manifolds, 80416

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Question 1

We got the following definition: Let $S, \bar{S} \subseteq \mathbb{R}^3$ be two surfaces and let $f : S \rightarrow \bar{S}$ be a smooth map. We say that f is locally conformal if for every p there exists a neighborhood $V \subseteq S$ such that

1. $f|_V : V \rightarrow f(V)$ is a diffeomorphism
2. f preserve angles between curves in V i.e. for every two curves $\alpha, \beta : I \rightarrow V$ such that there exists $t_0 \in I$ with $\alpha(t_0) = \beta(t_0)$ then

$$\frac{\langle \alpha'(t_0), \beta'(t_0) \rangle}{\|\alpha'(t_0)\| \|\beta'(t_0)\|} = \frac{\langle Df_{\alpha(t_0)}(\alpha'(t_0)), Df_{\beta(t_0)}(\beta'(t_0)) \rangle}{\|Df_{\alpha(t_0)}(\alpha'(t_0))\| \|Df_{\beta(t_0)}(\beta'(t_0))\|}$$

We will use the stereographic projection from exercise 5 to prove that for every $p_0 \in S^2$, $S^2 \setminus \{0\}$ is locally conformal to $\mathbb{R}^2$.

Proof: Without loss of generality we can assume $p_0 = (0, 0, 1)$ the North pole of the sphere $S^2 \subseteq \mathbb{R}^3$ and we defined $\pi : S^2 \setminus \{p_0\} \rightarrow \mathbb{R}^2$ to be the stereographic projection (which we saw that is diffeomorphism thus a local diffeomorphism everywhere) and we set its inverse $f = \pi^{-1} : \mathbb{R}^2 \rightarrow S^2 \setminus \{p_0\}$ both are given by

$$\pi(x, y, z) = \left(\frac{x}{1-z}, \frac{y}{1-z} \right)$$

$$f(u, v) = \left(\frac{2u}{u^2 + v^2 + 1}, \frac{2v}{u^2 + v^2 + 1}, \frac{u^2 + v^2 - 1}{u^2 + v^2 + 1} \right)$$

We notice that f is a smooth bijection with a smooth inverse (since all its components are rational functions with non-zero denominators). To satisfy condition 2 we must show that π preserve angles but it is sufficient to show that its inverse f is conformal (since the inverse of any conformal diffeomorphism is also a conformal as we saw in Complex Analysis; We can see it by definition with differentiation but it is kind of trivial since a map preserves angles going in one direction, reversing the mapping must preserve those exact same angles) so we compute f_u, f_v

$$f_u = \left(\frac{2(v^2 - u^2 + 1)}{(u^2 + v^2 + 1)^2}, \frac{-4uv}{(u^2 + v^2 + 1)^2}, \frac{4u}{(u^2 + v^2 + 1)^2} \right)$$

$$f_v = \left(\frac{-4uv}{(u^2 + v^2 + 1)^2}, \frac{2(u^2 - v^2 + 1)}{(u^2 + v^2 + 1)^2}, \frac{4v}{(u^2 + v^2 + 1)^2} \right)$$

We compute the coefficients of the First Fundamental Form E, F, G

$$E = \langle f_u, f_u \rangle = \frac{4}{(u^2 + v^2 + 1)^2}$$

$$G = \langle f_v, f_v \rangle = \frac{4}{(u^2 + v^2 + 1)^2}$$

$$F = \langle f_u, f_v \rangle = 0$$

Let $\lambda(u, v) = \frac{2}{u^2 + v^2 + 1}$ so we have $E = G = \lambda^2$ and $F = 0$.

For any tangent vector $w = a \frac{\partial}{\partial u} + b \frac{\partial}{\partial v} \in \mathbb{R}^2$ we have

$$\|Df(w)\|^2 = Ea^2 + 2Fab + Gb^2 = \lambda^2(a^2 + b^2) = \lambda^2\|w\|^2$$

By polarization, for any two vectors $w_1, w_2 \in T_q \mathbb{R}^2$, the inner product is

$$\langle Df(w_1), Df(w_2) \rangle = \lambda^2 \langle w_1, w_2 \rangle$$

If we look at two intersecting curves $\alpha, \beta \in \mathbb{R}^2$ where $w_1 = \alpha'(t_0)$, $w_2 = \beta'(t_0)$ The angle between their pushforwards $f \circ \alpha$ and $f \circ \beta$ on S^2 is

$$\frac{\langle Df(w_1), Df(w_2) \rangle}{\|Df(w_1)\| \|Df(w_2)\|} = \frac{\lambda^2 \langle w_1, w_2 \rangle}{\lambda \|w_1\| \lambda \|w_2\|} = \frac{\langle w_1, w_2 \rangle}{\|w_1\| \|w_2\|}$$

So f preserves angles therefore π its inverse must also preserve angle and is conformal map and we are done. $\square$

Question 2

Let $S, \bar{S} \subseteq \mathbb{R}^3$ be two surfaces and let $f : S \rightarrow \bar{S}$ be a diffeomorphism. For every curve $\gamma : [a, b] \rightarrow S$ we define the corresponding curve on $\bar{S}$, $\bar{\gamma} : [a, b] \rightarrow \bar{S}$ by $\bar{\gamma} = f \circ \gamma$.

We will prove that f is an isometry if and only if for every regular curve $\gamma : [a, b] \rightarrow S$, its arc-length parametrization $\varphi : [0, l(\gamma)] \rightarrow [a, b]$ is also arc-length parametrization for $\bar{\gamma}$.

Proof: $\Rightarrow$ Assume that f is an isometry.

By definition, an isometry preserves inner products on tangent spaces, therefore f preserves $\|Df_p(v)\| = \|v\|$.

Let $\gamma : [a, b] \rightarrow S$ be a regular curve and let $\varphi : [0, l(\gamma)] \rightarrow [a, b]$ be its arc-length parametrization so by definition $\alpha(s) = \gamma(\varphi(s))$ and $\|\alpha'(s)\| = 1$ for all $s \in [0, l(\gamma)]$.

Now for $\bar{\gamma}$ with $\bar{\alpha}(s) = \bar{\gamma}(\varphi(s)) = f(\alpha(s))$ by the chain rule

$$\alpha'(s) = Df_{\alpha(s)}(\alpha'(s))$$

Since f is isometry then on taking norm we will get that

$$\|\alpha'(s)\| = \|Df_{\alpha(s)}(\alpha'(s))\| = \|\alpha'(s)\| = 1$$

So $\bar{\gamma} \circ \varphi$ always has a norm of 1 then the parametrization φ is also the arc-length parametrization of $\bar{\gamma}$.

$\Leftarrow$ Assume that φ is always an arc-length parametrization for $\bar{\gamma}$.

So we have if $\alpha(s) = \gamma(\varphi(s))$ with $\|\alpha'(s)\| = 1$ then $\bar{\alpha}(s) = f(\alpha(s))$ also with $\|\bar{\alpha}'(s)\| = 1$ which means that for any unit vector $u \in T_p S$, its pushforward under f is also a unit vector.

Let $p \in S$ and $v \in T_p S$ be an arbitrary non-zero tangent vector so we can write $v = c \cdot u$ for $u = \frac{v}{\|v\|} > 0$ and $u = \frac{v}{\|v\|}$ a unit vector.

Since u is a unit vector, we can construct a regular curve in S through p with velocity u parameterized by arc length and by our assumption, its image under f will also have a velocity vector of norm 1 therefore

$$\|Df_p(u)\| = 1 \stackrel{\text{linearity}}{=} \|Df_p(c \cdot u)\| = c \cdot \|Df_p(u)\| = c \cdot 1 = \|v\|$$

Which means that Df_p preserves the norm of all tangent vectors.

TO show that Df_p is isometry we need to show that it preserves inner products

$$\langle v, w \rangle = \frac{1}{4}(\|v + w\|^2 - \|v - w\|^2)$$

Following

$$\begin{aligned} \langle Df_p(v), Df_p(w) \rangle &= \frac{1}{4} \left(\|Df_p(v) + Df_p(w)\|^2 - \|Df_p(v) - Df_p(w)\|^2 \right) \\ &= \frac{1}{4} \left(\|Df_p(v + w)\|^2 - \|Df_p(v - w)\|^2 \right) \\ &\stackrel{(\star)}{=} \frac{1}{4} (\|v + w\|^2 - \|v - w\|^2) \\ &= \langle v, w \rangle \end{aligned}$$

Where $(\star)$ is due to the fact that it preserves norms.

Therefore f is isometry since it is a diffeomorphism and preserves inner products on all tangent spaces. $\square$